

- ⑧ Train-test-split

- ⑤ Model Training

- ## 10 Model Evaluation

- ⑪ hyper parametric tuning

- 12 Deployment
Monitor

df.loc[row col]

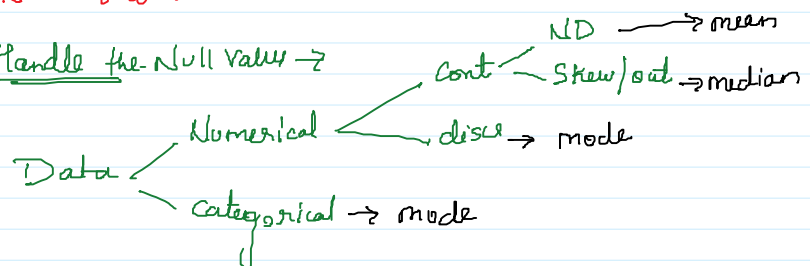
df.loc[df.max().index(),]

1) Detect & Handle the Null values

Detect

- info()
- describe()
- isnull()

Handle the Null Value →



Age

35

Not

40

Nov

65

75

II Detect & Handle the Outliers.

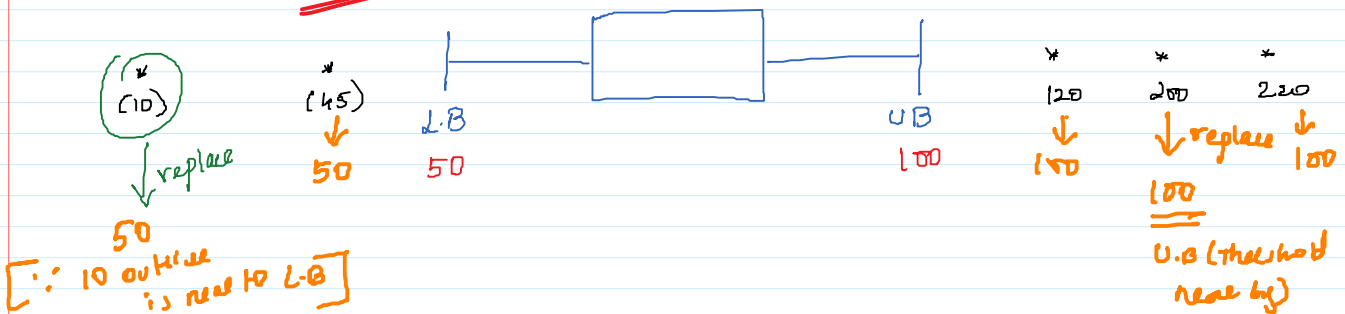
- a) Detect $\rightarrow$ ^{methods}
- ① boxplot
 - ② IQR $\rightarrow \left. \begin{array}{l} \text{val} < \text{Low-bound} \\ \text{val} > \text{Upper-bound} \end{array} \right\} \rightarrow \text{define Outliers}$
 - ③ Zscore
 - ④ Empirical rule $\rightarrow$ Value is beyond the limit $\mu - 3\sigma$ to $\mu + 3\sigma$

b) Handle the outliers.

a) 2-3% drop/delete the outliers

b) Capping method $\rightarrow$ Outlier values are replaced with Nearby Threshold value (lower bound/upper bound)

Winzonisation



Scaling is the process of transforming numerical data so that features are on a similar scale

Scaling :- Convert all numerical data into one form

- ① MinMax Scales $\rightarrow 0$ to 1 $\rightarrow$ deep Algo, Anni, Image Extract
- ② Standard Scales $\rightarrow -3$ to $+3$ $\rightarrow$ Distance based Algo.
- ③ Robust Scales $\rightarrow$ median, IQR $\rightarrow$ Handles outliers

① Null values

② Duplication

③ Outliers

④ Transformation

⑤ Encoding

⑥ Scaling

Min-max

$$\text{Min-max} = \frac{X - X_{\min}}{X_{\max} - X_{\min}}$$

Standard Scales (Zscore)

$$Z_{\text{score}} = \frac{X - \text{mean}}{\sigma}$$

Robust Scales

$$R = \frac{X - \text{median}}{IQR}$$

Encoding :- Handling the categorical data

→ Categorical data is converted to Numerical data

Eg:- Female → 0
male → 1

Types of Categorical data

- Nominal data → doesn't have order
- Ordinal data → has order

① OHE → One Hot Encoding Technique
↳ Nominal

<u>Gender</u>	Gender_Female	Gender_Male
Male	0	1
Male	0	1
Female	1	0
Male	0	1
Female	1	0

<u>Colour</u>	Colour_Red	Colour_Green	Colour_Blue
Red	1	0	0
Green	0	1	0
Red	1	0	0
Blue	0	0	1
Green	0	1	0
Blue	0	0	1

⇒

2) Label encode

<u>Gender</u>	<u>Gender-label</u>
Female	0
Male	1
Female	0
Male	1

(Alphabetic)
 Unique → Female → 0
 Male → 1

Target val — X
 ↓
 ML autom

Colours

Red
 Green
 blue

Alph → Blue — 0
 Green — 1
 Red — 2

3) Ordinal encode → Ordinal data

Size → S, M, L, XL, XXL → define
 ↓ ↓ ↓ ↓ ↓
 0 1 2 3 4

Eg:- Principal → 2
 Student → 0
 Teacher → 1

4) frequency encode

Colours
 ✓ Red → 3
 Green → 4
 ✓ Red → 3
 Blue → 2
 Green → 4
 ✓ Red → 3
 Blue → 2
 Green → 3
 Green → 3

freq
 Red → 3
 Green → 4
 Blue → 2

5) Target encode

place
 ✓ Bombay → 3
 Bay → 1
 Delhi → 2
 Delhi → 2
 ✓ Bombay → 3
 ✓ Bombay → 3

Nominal - OHE, label enc X

Encoding
 Categorical
 data

Feature → Handle the categorical

Target → Not handling categorical

Species

Feature → model

Feature → model

Species

Setosa → 0
 Vers → 1
 Virg → 2 } label encode
 automati

Colour → label encoder

Red → 2

Green → 1

blue → 0

model $\frac{Q}{2} \frac{1}{2} \frac{0}{0} \xrightarrow{\text{label}} \text{relat} \rightarrow \text{ranking}$

Feature
 OHE ✓

0 0 1
 0 1 0
 1 0 0 } No relat

Summarize :- Data Transformation

- Normalization
- Encoding
- Scaling

transform → Skew data → NP

→ >0 → R
 <0 → L

moderate
 No trans

More
 Skewness
 R L R+L

log x

cube

power

log x+1

① +ve, -ve

② zero?

③ More skew

Scaling / Feature ✓ → Continuous data

✓ Target

iris → SL PL PW PL → Scaling
same range of values

Age | salary
25 | 50000 → Scaling

* distance-based algo → Scaling is must

↳ Standard

Neural net
↳ 0 to 1
min max

* Robust Scales

↳ mainly when outliers present

Encoding Categorical

- ONE → 1 0 0 → Vectors
- label → Alphabet → 0 to n
- Ordinal → Based on priority / ranking
- Freq → Based on freq of that value

④ KNN → k Nearest Neighbour

→ Supervised M.L

→ labelled data

→ principle : 1) Distance based algo
2) prob based algorithm

→ Lazy algo

→ It works only / Active only when there is test data

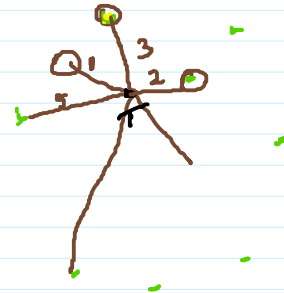
→ No training the data

→ k → no. of neighbours

- No training the data
- $K \rightarrow$ No. of Neighbours
- $K=3$
- Regression + classification



→ Training data
→ data
 $K=3$



$$Reg = \frac{5+4+6}{3} = \text{avg of the values}$$

Classifi = $K \rightarrow$ odd No. 3, 5, 7

$K=3$

$K=4$
y * 4
N * 2

highest prob of class $\Rightarrow$ Yes

Yes $\Rightarrow 2$

No $\Rightarrow 1$

Predict from model $\rightarrow$ Yes



SAMPLING TECH.

→ To make Dataset Balance

→ Classification

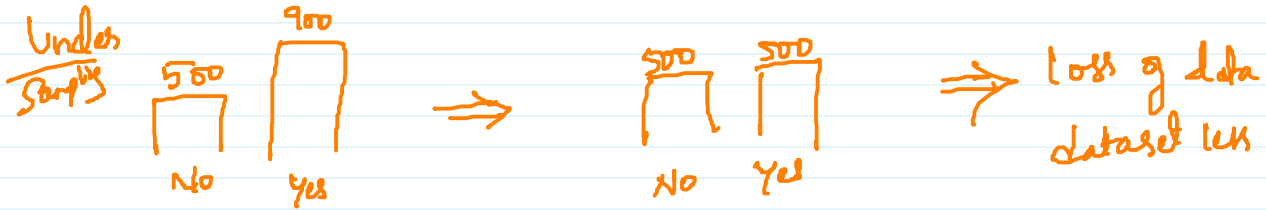
Yes $\rightarrow 900$ records
No $\rightarrow 500$ Records } Imbalanced

Yes $\rightarrow 900$
No $\rightarrow 900$ } Balanced

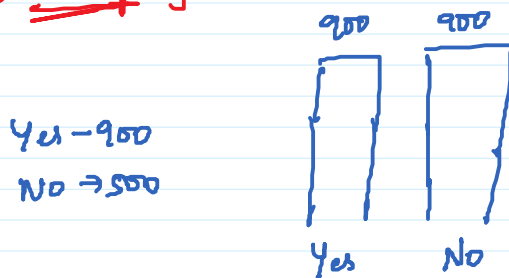
→ Imbalanced $\rightarrow$ Model is biased towards more records

Tech $\rightarrow$ Undersampling $\rightarrow$

OverSampling $\rightarrow$
 SMOTE $\rightarrow$



OverSampling



$\rightarrow$ duplicate values (400)

$\rightarrow$ Model $\rightarrow$ overfitting

Same data is trained multiple

Model 1

$\rightarrow$ Training $\Rightarrow$ 95% ✓
 data

$\rightarrow$ Test $\Rightarrow$ 60% ✗

Overfitting

Model 2

Train: Acc = 82% ✓

Test Acc = 80% ✓

Generalised Model

Model 3

Train = 45% ✗

Test = 3% ✗

Underfitting

SMOTE $\rightarrow$ Synthetic Minority OverSampling Technique

